



OPEN ACCESS

EDITED BY
Akinkunmi Paul Okekunle,
University of Ibadan, Nigeria

REVIEWED BY
Hernan Bayona,
Fundacion Valle del Lili, Colombia
Alvaro Soto,
University of La Frontera, Chile

*CORRESPONDENCE
Christopher A. Alarcon-Ruiz
✉ christoper.alarconr20@gmail.com

RECEIVED 03 July 2026
REVISED 11 August 2026
ACCEPTED 18 August 2026
PUBLISHED 28 August 2026

CITATION
Alarcon-Ruiz CA, Gutierrez-Arratia J,
Concha-Querevalú L, Barrientos-
Ventura E, Moreno-Arias J, Otiniano-
Sifuentes R, Ecos R, Zunt JR and
Miranda J.J (2026) Ischemic stroke in
Peru: a scoping review of trends,
outcomes, and evidence gaps.
Front. Epidemiol. 6:1926861.
doi: 10.3389/fepid.2026.1926861

COPYRIGHT
© 2026 Alarcon-Ruiz, Gutierrez-Arratia,
Concha-Querevalú, Barrientos-Ventura,
Moreno-Arias, Otiniano-Sifuentes, Ecos,
Zunt and Miranda. This is an open-
access article distributed under the
terms of the [Creative Commons
Attribution License \(CC BY\)](https://creativecommons.org/licenses/by/4.0/). The use,
distribution or reproduction in other
forums is permitted, provided the
original author(s) and the copyright
owner(s) are credited and that the
original publication in this journal is
cited, in accordance with accepted
academic practice. No use, distribution
or reproduction is permitted which does
not comply with these terms.

Ischemic stroke in Peru: a scoping review of trends, outcomes, and evidence gaps

Christopher A. Alarcon-Ruiz^{1,2*}, Jesus Gutierrez-Arratia¹,
Lessly Concha-Querevalú¹, Erick Barrientos-Ventura³,
Jacqueline Moreno-Arias⁴, Ricardo Otiniano-Sifuentes⁴,
Rosa Ecos⁴, Joseph R. Zunt⁵ and J. Jaime Miranda^{6,7}

¹Instituto Nacional de Ciencias Neurológicas, Lima, Peru, ²Unidad de Investigación Para la Generación y Síntesis de Evidencias en Salud, Universidad San Ignacio de Loyola, Lima, Peru, ³Facultad de Ciencias de la Salud, Universidad San Ignacio de Loyola, Lima, Peru, ⁴Departamento de Enfermedades Neurovasculares y Metabólicas, Instituto Nacional de Ciencias Neurológicas, Lima, Peru, ⁵Departments of Neurology, Global Health, Medicine (Infectious Diseases), and Epidemiology, University of Washington, Seattle, WA, United States, ⁶Sydney School of Public Health, Faculty of Medicine and Health, The University of Sydney, Sydney, NSW, Australia, ⁷CRONICAS Centre of Excellence in Chronic Diseases, Universidad Peruana Cayetano Heredia, Lima, Peru

Background: Peru faces a growing mismatch between the increasing burden of ischemic stroke and the health system's capacity to respond.

Aim: We aimed to systematically map the primary research literature on ischemic stroke in Peru using a socio-ecological framework, characterize the current evidence base, quantify key epidemiological outcomes, and identify critical knowledge gaps to inform a national research agenda.

Methods: We conducted a scoping review using the SPIDER framework and searched four databases through April 2025. We included original primary studies of Peruvian participants of any age with ischemic stroke, including observational, interventional, qualitative designs, and case reports. Study selection and data extraction were performed in duplicate. Findings were narratively synthesized across socio-ecological levels (individual, interpersonal, organizational, community, and policy/environmental). Key epidemiological outcomes were pooled using random-effects meta-analyses.

Results: Of 1210 records, 119 studies were included (131,424 participants), with 75% conducted in Lima. Ischemic stroke represented 62% of all stroke cases; 44% were women. Overall mortality was 12%. Large-vessel atherosclerosis (27%) and cardioembolic (24%) were common etiologies, but undetermined causes remained the highest (30%). The pooled thrombolysis rate was 2.9%, with no reported access to mechanical thrombectomy in observational studies. Qualitative studies identified barriers to timely care, poor guideline adherence, caregiver burden, and centralization of services. Quality-improvement efforts showed limited impact. Community prevention and digital innovations were tested in small pilots without scale-up. Policy-level evaluation was scarce.

Conclusion: Research regarding stroke in Peru has increased but remains geographically centralized and structurally misaligned with the national burden. Major gaps persist in diagnostic capacity, implementation research, rehabilitation, caregiver support, and policy evaluation. A nationally coordinated and regional distributed research and system-level strengthening of the stroke program are urgently needed.

Systematic Review Registration: (<https://doi.org/10.17605/OSF.IO/7WYX8>).

KEYWORDS

epidemiology, ischemic stroke, outcome, prevention, public policy

Introduction

Stroke remains among the leading causes of death and long-term disability worldwide, with associated burden rising substantially over the past three decades (1, 2). In 2021, an estimated 11.9 million incident strokes occurred worldwide, 65% of which were ischemic stroke (1). Recent population-based estimates report an age-standardized incidence of ischemic stroke in Peru of approximately 1.65 per 10,000 persons yearly and a marked altitudinal gradient, highlighting Peru's unique geographic and demographic determinants of risk and care access (3). Ischemic stroke has different pathophysiological mechanisms, acute management protocols, and secondary prevention strategies, that often differ significantly from those of hemorrhagic stroke (4).

The burden of ischemic stroke is disproportionately concentrated in low- and middle-income countries, which account for approximately 86% of ischemic stroke deaths and 87% of disability-adjusted life years (5). In Latin America and the Caribbean, rapid demographic and epidemiological transitions, layered onto pre-existing health-system limitations, continue to maintain high morbidity of ischemic stroke (6). These limitations including restricted thrombectomy capacity, nascent telemedicine networks, and limited access to timely rehabilitation, highlight the need for context-specific implementation research (7, 8).

Peru exemplifies these challenges in a unique epidemiological setting. Local data show a critical gap in ischemic stroke care: at the national tertiary referral neurology institute in Lima, less than 2% of patients with ischemic stroke received intravenous alteplase between 2009 and 2016, a pattern driven largely by delayed presentation and system barriers (9). Physical and human health resources are highly centralized in urban settings and care pathways are fragmented (10, 11). Access to advanced imaging and mechanical thrombectomy is limited outside the capital and largely concentrated in tertiary centers, and rehabilitation services are underdeveloped (11, 12).

Unlike global or regional analyses, country-level evidence mapping in a lower-middle-income setting allows identification of where the research data is structurally misaligned with the actual disease burden and geographic distribution of care (13). The mismatch between a growing disease incidence and burden (3) and the country's limited capacity to respond (12, 14) demonstrates the urgency of a targeted research agenda in Peru. This scoping review has a single primary objective: to systematically map the existing primary literature on ischemic stroke in Peru using a socio-ecological framework, in order to characterize the evidence base, quantify key epidemiological outcomes where feasible, and identify the most critical knowledge gaps that should guide a national research agenda. This framework was selected because we aimed to assess the determinants of ischemic stroke prevention and care extend beyond the individual clinical level to interpersonal, organizational, community, and policy/environmental factors.

Methods

Study design and protocol

This study reports a scoping review in accordance with the PRISMA checklist for Scoping Reviews (Supplementary Table S1). The protocol was previously developed by the authors and is available on OSF.io (<https://doi.org/10.17605/OSF.IO/7WYX8>).

We followed a mixed quantitative and qualitative approach utilizing the SPIDER framework (15) to guide the scoping and synthesis of evidence. Within this framework, our Sample of interest was the Peruvian population; the Phenomenon of Interest was specifically the ischemic stroke; the eligible Design included published literature of any research design; and the Research Type encompassed primary quantitative or qualitative research data, including case reports. The resulting body of evidence was subjected to Evaluation following a socio-ecological approach, which structured the synthesis around five key categories for determinants: 1) Individual, 2) Interpersonal, 3) Organizational, 4) Community, and 5) Policy and environment (16). Unlike other frameworks focused primarily on intervention planning, such as PRECEDE-PROCEED (17), or on sequential clinical care pathways (18), the socio-ecological approach allows heterogeneous evidence to be mapped across multiple levels of the health system, consistent with the objective of identifying structural gaps in Peru.

Search strategy and selection criteria

We systematically searched in Scopus, Web of Science Core Collection, Scielo via Web of Science, and PubMed/Medline databases, using the terms of "Stroke", "Ischemic cerebrovascular event", "Acute ischemic attack", "Apoplexy", "Infarct", and "Peru", to ensure maximum initial comprehensiveness. Detailed search strategies for each database can be found in Supplementary Table S2. The search date was carried out on January 17th, 2024, and was updated only in Scopus database on April 4th, 2025, due to methodological efficiency, avoiding duplicated efforts, and its superior coverage for retrieving the highest number of results, usually overlapped, compared with PubMed/Medline and Web of Science.

We included original studies conducted in human participants of any age diagnosed with ischemic stroke or reports ischemic stroke as an outcome. Eligible study designs included observational studies, interventional studies, and qualitative research. The latter involved patients, caregivers, or healthcare professionals engaged in the care of individuals with any type of stroke. Also, we included studies that proposed novel rehabilitation interventions for patients with stroke, ischemic or hemorrhagic, because their disability is similar. Only studies that reported specific information on persons within the Peruvian territory were included. Availability or reporting of neuroimaging was not a prespecified eligibility criterion.

Consistently with a scoping review framework that explicitly values breadth of coverage, case reports and case series may present uncommon or complex stroke etiologies that would

otherwise be invisible in aggregate epidemiological datasets. While conference abstracts published in peer-reviewed scientific journals sometimes represent the only primary publications available for describing new technological interventions or small pilot interventional studies. Thus, we included case reports and case series that describe uncommon etiologies or treatments of ischemic stroke, as well as conference abstracts published in peer-reviewed scientific journals for novel rehabilitation studies and pilot interventional studies.

In case of studies that included both ischemic and hemorrhagic stroke, those were included if they presented separate data for only ischemic stroke. The following types of studies were excluded from the review: studies focused on global ischemic cerebrovascular disease (e.g., post-anoxic encephalopathy or global cerebral ischemia), editorials, letters to the editor without original results, and reviews that did not report primary data. Studies focused on cerebral venous thrombosis or venous infarction were also excluded because these conditions have distinct pathophysiological mechanisms, diagnostic pathways, and acute and secondary treatment strategies compared with arterial ischemic stroke. The review therefore focused specifically on arterial ischemic stroke.

Study selection and data extraction

Initially, the results from each search were exported to the free online Rayyan QCRI software (<https://rayyan.qcri.org>; Qatar Computing Research Institute, Qatar Foundation, Qatar) and duplicated references were eliminated. Then, two independent researchers evaluated the titles and abstracts and then conducted a full-text review as a secondary step (CAA-R and JG-A). The discrepancies were resolved by consensus in both processes.

Data from the included studies were independently extracted by the six researchers paired in three independent groups (CAA-R and JG-A, RO-S and JM-A, RE and LC-Q) using a data collection form in Microsoft Excel (Microsoft, Washington, USA) designed for each study design. For case reports, we collected demographic (sex and age) and clinical data [symptoms, vascular territory, TOAST etiology reported by the article and assessed by the reviewer [large artery atherosclerosis, cardioembolic, small vessel, other determined, or undetermined] (19), treatment, and follow-up]. For observational studies, we collected population characteristics (location, time period, number of included subjects, and number of subjects for each etiology following the TOAST classification and as reported by individual studies, when available), magnetic resonance imaging (MRI) or advance perfusion imaging use, epidemiological outcomes (prevalence, incidence, mortality, disability), and clinical characteristics (arrival time to hospital, etiology, reperfusion rates, COVID-19 related information, associated factors or outcomes). We did not find studies reporting a different classification system for ischemic stroke rather than TOAST criteria.

For qualitative studies, we collected study population and different subjects' reports (healthcare access, rehabilitation, psychological impact, costs, personal experiences, and COVID-10 related information). Finally, for intervention studies we collected population information, type and details for

intervention, and main results. All data collected were retrieved by two independent investigators (CAA-R and JG-A, RO-S and JM-A, RE and LC-Q), and in case of discrepancies, they were solved by consensus.

Synthesis of results

Most findings were summarized narratively following a socio-ecological health approach, dividing the results into five levels: Individual, Interpersonal, Organizational, Community, and Policy and environment. Studies were classified according to the primary level at which the determinant or barrier operated; when findings overlapped across levels, classification was resolved through reviewer discussion and consensus. We also produced descriptive frequency analyses of study design, year of publication, and geographic setting, using a curated dataset derived from the extraction table. For quantitative synthesis we prespecified several outcomes: (i) proportion of ischemic stroke among all cerebrovascular events, (ii) proportion of ischemic stroke within cohorts recruited for another disease, (iii) proportion of women, (iv) mortality rate, (v) thrombolysis rate, and (vi) etiology classification among ischemic stroke cases. We pooled outcomes when at least three studies reported compatible numerators and denominators from the same source population after cleaning and harmonization. When multiple reports referred to the same cohort, we retained the record with the largest valid denominator or the most complete information and removed obvious duplicates. If a report provided several time periods, we used study year pairs only when period-specific denominators were explicit; otherwise, a single overall estimate was retained. Studies with incompatible definitions or unresolved inconsistencies were excluded from that synthesis but retained for the narrative description. All quantitative syntheses were implemented in R using the *meta* and *metafor* packages, with modeling details provided in the Statistical analysis section.

Prespecified outcomes

We synthesized several key outcomes related to ischemic stroke in Peru. Primary metrics included: i) proportion of ischemic stroke among all cerebrovascular events, calculated using the sum of ischemic and hemorrhagic strokes within the same source population as the denominator. To capture ischemic stroke burden as a complicating condition in non-stroke-primary study populations, we evaluated ii) proportion of ischemic stroke within non-stroke cohorts recruited for other diseases or conditions, using the total size of the non-stroke cohort as the denominator. Studies that only enrolled stroke patients were excluded from this specific synthesis.

To characterize the patient population and acute management, we also calculated iii) proportion of women among all ischemic stroke cases, and iv) proportion of deaths (both intrahospital or after discharge deaths) among all ischemic stroke cases. Regarding acute care access, we quantified v) proportion of thrombolysis utilization among all ischemic stroke cases. Finally, we aimed to map the underlying causes by determining vi)

proportion of each stroke etiology (large artery atherosclerosis, cardioembolic, small vessel, other determined, or undetermined) among the total number of ischemic stroke cases.

Statistical analysis

Proportions were modeled with *metaprop* on the logit scale (PLOGIT) and were backtransformed for presentation. Random-effects models were the primary approach. Between-study variance (τ^2) was estimated with restricted maximum likelihood, and confidence intervals used the Hartung-Knapp adjustment. To enable computation of overall estimated proportion, in studies with zero-event cells, a continuity correction of 0.5 was applied uniformly, when required. We reported I^2 , τ^2 , and a 95 percent confidence interval for each proportion metanalysis. Common-effect estimates were provided as a reference.

Frequency analysis and graphics

A consolidated dataset captured publication year, study design, geographic distribution, and center type. Annual counts by design were computed and plotted as stacked bars to show temporal trends. Geographic distribution and the share of single-center and multicenter studies were summarized from the same curated dataset to maintain internal consistency across the study characteristics table and figures.

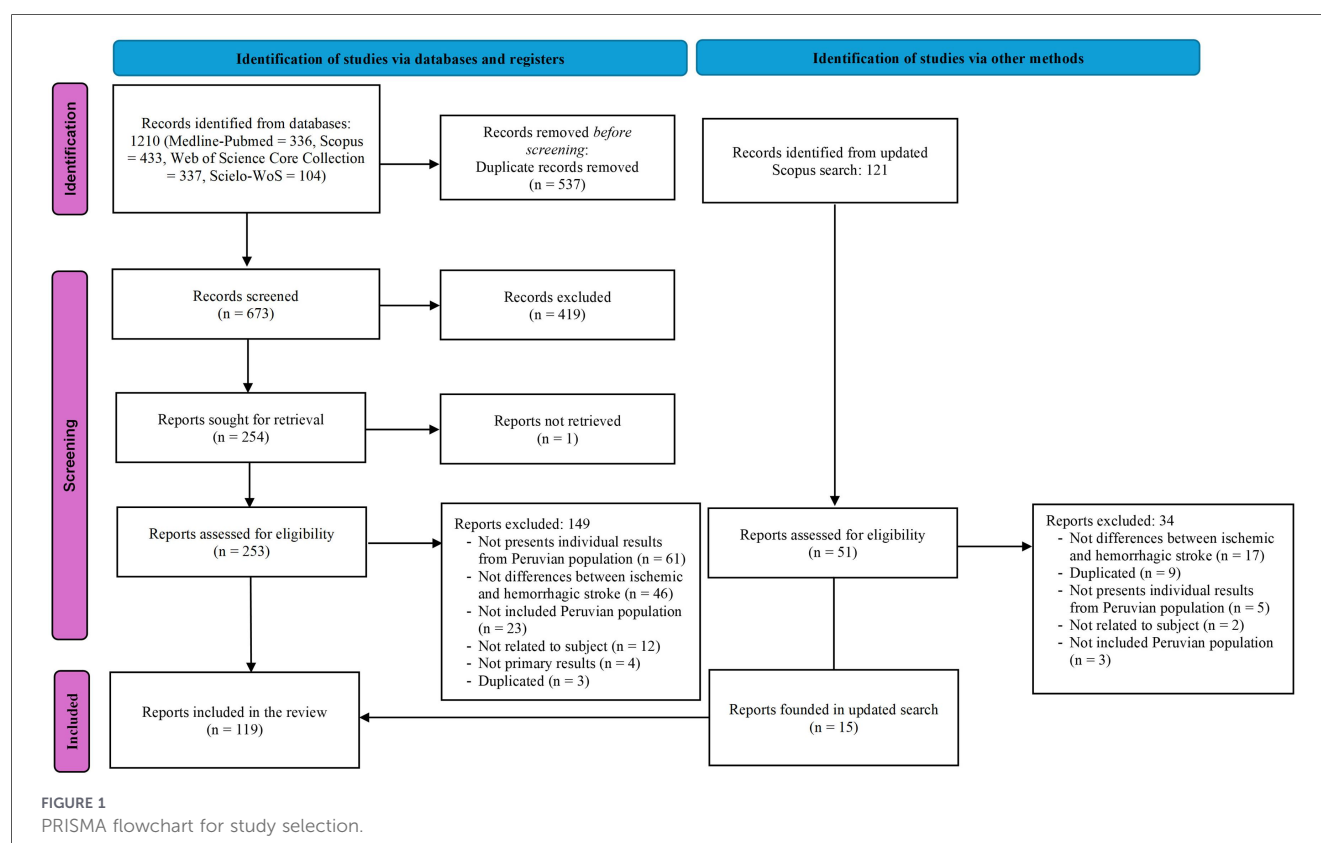
Software and reproducibility

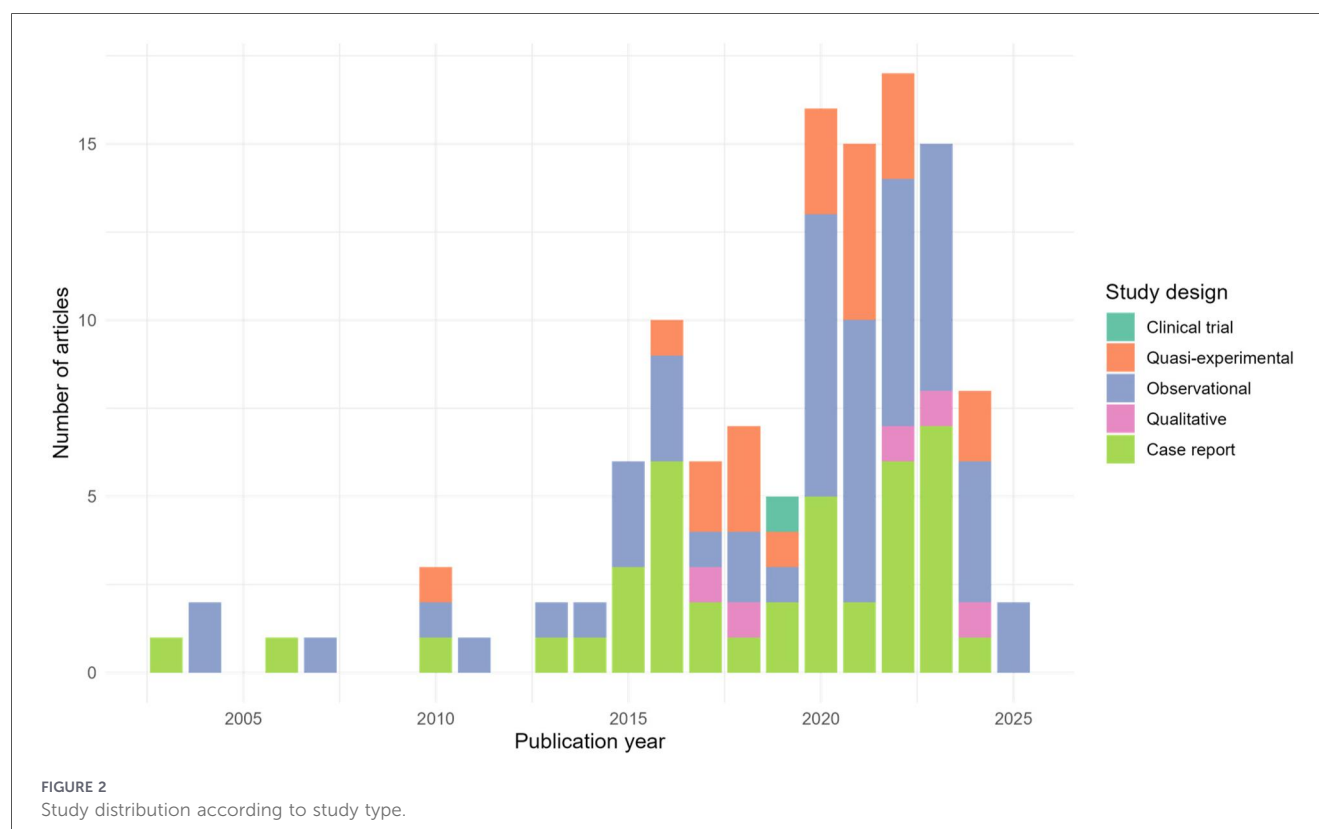
All quantitative syntheses were implemented in R using the meta and metafor packages. Data handling and figure generation used tidyverse utilities as well as readxl, janitor, stringr, stringi, and openxlsx for spreadsheet input and output. Forest plots reflect the logit transformation, restricted maximum likelihood for τ^2 , the Hartung-Knapp adjustment, and 95 percent confidence intervals. Curated included and excluded datasets, pooled summaries, influence tables, and figures were exported programmatically and are available upon request together with the scripts and output directories.

Results

Study selection and characteristics

We identified 1,210 records through database search, and after removing duplicates, 673 records were screened. Two hundred fifty-three records were evaluated in full text, and 149 records were excluded. We identified five randomized clinical trials for ischemic stroke prevention, two for acute treatment, one for secondary prevention and one for rehabilitation, but all were excluded because they did not present independent results of the Peruvian population (reasons provided in [Supplementary Table S3](#)). Following an updated search in Scopus, 119 articles were included in this scoping review ([Figure 1](#)). The final sample comprised 53 observational studies ($n = 131\ 370$ study participants), 38 case reports documents ($n = 54$ subjects), 21





quasi-experimental studies, five qualitative studies, and one cluster randomized clinical trial. Among all, 75.4% were from Lima. Most of these studies were published during or after 2020 (Figure 2).

Individual level: case reports

The included cases represent a substantial portion of the available primary literature on ischemic stroke in Peru, particularly for uncommon etiologies. Reports were retrospectively collected and were published between 2003 and 2024. Among the 54 reported individuals, the median age was 45 years old with range from 5 to 91 years old, with females as 53%. Cases were reported mostly from institutions in Lima (87%), while two cases were from Arequipa and two from La Libertad, and one case from Cajamarca, Ica and Piura, each (Figure 3B). Most of the cases (80%) were ischemic strokes of anterior circulation. Undetermined etiology was the most frequent diagnosis (48%), followed by other determined etiology (35%) and cardioembolic etiology (6%). However, there were 7 cases where the etiology mentioned by the report were cardioembolic or other determined etiology, but after the critical revision, we could not find enough evidence to confirm those etiologies. The most common symptoms were motor deficit (83%), cognitive or behavioral impairment (35%), headache (15%), and epileptic seizures (13%).

Individual level: epidemiological studies

The compiled database of 53 observational studies of ischemic stroke in Peru demonstrated substantial heterogeneity across

populations and reported outcomes, limiting direct comparability but offering a broad portrait of the national research landscape. The aims of these studies were as follows.

a) studies where ischemic stroke was the primary recruitment criterion (34%); (b) studies where ischemic and hemorrhagic stroke patients were included (20%); and (c) studies where ischemic stroke was a secondary or incidental outcome within a non-stroke cohort (22%) or in a general population (24%). Among this last general population group, most studies used nationwide databases.

Among 53 observational studies, most were from a single center (79%), conducted in Lima (76%), and recruited participants between 2012 and 2022 (Table 1). The representation of rural, high-altitude Andean or Amazonian settings was almost null (Figure 3A) (20–22). Many of these studies reported different epidemiological estimates (frequency and mortality), but methods, sample size, denominators, and study periods varied widely. The pooled proportion of ischemic stroke was 65% among all ischemic and hemorrhagic stroke cases ($n=36,513$) (Supplementary Figure S1), however, there was not specific data regarding sex and age among most of these studies. Across studies with available data, women represented 44% of ischemic stroke cases ($n=5,821$) (Supplementary Figure S2); mean age among all ischemic stroke cases ($n=3,396$) was 66.9 years old (Supplementary Figure S3); and 8% of patients with other non-stroke conditions like COVID-19, cardiovascular diseases, and systemic infections ($n=104,693$) experienced an ischemic stroke (Supplementary Figure S4). Regarding etiology, large-vessel atherosclerosis was the most frequent determined etiology (25%), followed by cardioembolic (23%), small-vessel disease (23%), and other

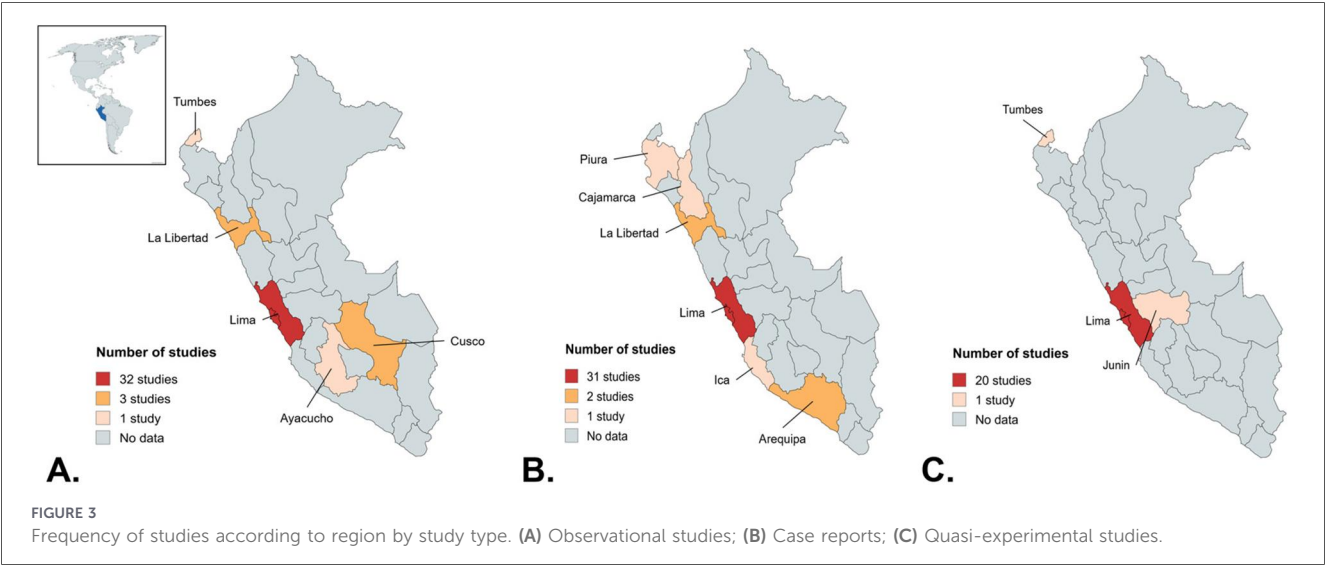


TABLE 1 Observational studies characteristics regarding ischemic stroke in Peru.

Variable	Categories	Total	General population	Ischemic or hemorrhagic stroke	Only ischemic stroke	Non-stroke cohorts (Other pathology)
Centers	One center	33	1	8	13	11
	Multicentric	9	2	1	5	1
	Nation-wide secondary data	11	8	2	0	1
Region	Lima	32	1	7	14	10
	Cusco	3	0	0	1	2
	La Libertad	3	0	1	2	0
	Tumbes	1	1	0	0	0
	Ayacucho	1	0	1	0	0
Study period	Before 2012 (Pre-thrombolysis)	17	2	5	5	5
	2012	8	0	1	3	4
	2013	7	0	1	3	3
	2014	7	0	1	3	3
	2015	12	3	1	5	3
	2016	14	3	3	5	3
	2017	11	2	3	2	4
	2018	9	1	2	3	3
	2019	10	2	2	3	3
	2020	12	1	2	4	5
	2021	10	2	1	3	4
	2022	5	1	0	2	2

determined etiologies (3%). However, the pooled proportion of undetermined etiology was the highest at 30% (Supplementary Figures S5–S9). Only two observational studies from Lima reported the number of patients with MRI use (23, 24), and no study reported the use of advanced perfusion imaging.

Among patients with ischemic stroke, overall mortality was 12% (Supplementary Figure S10), of which most occurred while hospitalized. Only one study reported one-year mortality (25). There was no information regarding mortality stratified by imaging modality, infarct volume, or imaging-defined stroke severity. Disability and other clinical outcomes were reported inconsistently across studies, but several series highlighted a high burden of post-stroke disability (9, 22, 26–31) Risk-factor data emphasized hypertension, dyslipidemia, cardiovascular

disease, hormonal contraceptive, and aging as recurrent contributors for ischemic stroke development (32, 33). Additionally, there were reports of some risk factors for worse outcomes among ischemic stroke patients, such as inflammatory biomarkers (high C-reactive protein, high white blood count, and low acid uric levels) (30, 34, 35) and intrahospital complications (hydrocephalus, hemorrhagic conversion, infections) (33, 34, 36). Compared to hemorrhagic strokes, patients with ischemic stroke had lower mortality and shorter hospital stays (29, 37, 38). Finally, a few studies reported the association between ischemic stroke with depression (39) and cognitive impairment or dementia (40, 41).

Individual level: feasibility of computer-based innovations

Eighteen studies proposed innovative and accessible robotic or computer-assisted solutions for post-stroke patients, mostly from Lima (Figure 3C). These included eight reports of brain-computer interphase validated in post-stroke patients to be used in smart-homes or other adaptive software for their lifestyle activities (42–49). Eight additional studies reported initial validation of novel interventions for upper limb rehabilitation using exoskeletons (50–54), virtual reality (55, 56), and an orthosis (57). While other studies described the telerehabilitation communication system for post-stroke patients (58), and a novel algorithm for gait assessment in post-stroke patients with tactile impairment in lower legs (59). Most of them were software presentations with small sample size of patients with stroke and healthy controls to pilot them. Despite these projects representing a promising advancement in healthcare innovations, they remained at a very early stage of development with no assessment within a large-scale population.

Interpersonal level: caregivers and healthcare providers

Five qualitative studies reported the challenges of caregivers and healthcare providers caring for post-stroke patients, all from Lima. Caregivers' worries and fears were linked to their patients' new disabilities and the presence of invasive instruments such as nasogastric tube (60). Additionally, caregivers faced reduced social activity and unexpected financial burdens (61) including hiring private services to overcome care gaps (62). This new situation causes increase caregiver burden (62) and emotional stress and depressive symptoms, which increase their needs to see a mental-health professional (61).

Most healthcare providers lack correct training for stroke rehabilitation (63). Caregivers reported receiving only oral, not written, information about rehabilitation with no practical training in post-stroke care tasks after discharge (64). While healthcare providers indicated that only nurses were responsible for delivering this information (63). However, interventions with interprofessional education may increase the knowledge and confidence of healthcare providers and caregivers for care of post-stroke patients (64).

Organizational level: access to healthcare in stroke

Patients described multiple barriers to timely emergency care, including poor awareness of stroke (64) and limited understanding of how the health system works (64). Five observational studies reported the time from symptom onset to hospital arrival in patients with ischemic stroke: In patients who received thrombolysis, mean time was 134.2 min (9, 36), while in other studies, arrival times ranged between 3 h to 7 days (35, 65, 66). Among ten studies reporting thrombolysis rates, overall frequency was 2.9% (Supplementary Figure S11). No study reported data regarding indications for or access to mechanical thrombectomy, although the procedure is performed in few tertiary public and private centers in Peru (67, 68).

During hospitalization, caregivers highlighted communication challenges with healthcare providers, such as nonfunctioning phone lines, limited information sharing, inconsistent explanations between providers, and at times impolite interactions (63). In addition, after hospital discharge, caregivers reported feeling underprepared to manage care of the patients and noted geographic disparities in access to dysphagia services (60), and barriers to obtaining a referral for specialized stroke management using the formal referral pathway (63).

Organizational level: adherence to recommendations

One local study reported mean adherence to AHA/ASA stroke performance measures of 47%. Only five of 15 AHA/ASA stroke performance measures were achieved at a level of 85% or higher: antithrombotic treatment at discharge, antithrombotic treatment within 2 days of hospitalization, discharge on statins, rehabilitation assessment and cardiac monitoring within 2 h of arrival and continuing for 24 h during hospitalization (26). In this context, a cluster randomized clinical trial evaluated whether a multifaceted quality improvement intervention could increase adherence to evidence-based therapies for the management of patients with acute ischemic stroke and transient ischemic attack in a hospital in Lima. However, the intervention did not achieve significant improvements in overall adherence (69).

Community level: COVID-19

During the COVID-19 pandemic, there was an increase in published studies about ischemic stroke in Peru. There were five case reports of ischemic stroke in the context of COVID-19 infection in children, pregnant women, and adults (70–72). Across observational studies, several findings highlighted the relationship between stroke care and the COVID-19 pandemic in Peru. One analysis documented a relative reduction of 46.9 per 1,000 inhabitants in overall stroke volume, with an even sharper decline of 70.8 per 1,000 inhabitants in intravenous tPA procedures during the pandemic (73). Other comparisons of pre- and post-pandemic admissions found no statistically

significant differences for acute ischemic stroke ($p=0.34$) or transient ischemic attack ($p=0.46$) (74). Patients with concomitant COVID-19 infection were noted to have higher baseline glucose, elevated neutrophil counts, reduced lymphocyte counts, and an increased neutrophil-to-lymphocyte ratio, alongside higher NIHSS scores at presentation (35). This might explain why stroke in the setting of COVID-19 was associated with high mortality and low rates of functional independence among survivors (24).

Limitations in access to healthcare were exacerbated during this period. During the lockdown, patients and their caregivers had limited mobilization around the city to arrive at a hospital. Also, they had trouble understanding the reorganization of stroke healthcare. Communication between caregivers and medical staff was reduced to fragmented telephone calls, no formal training was provided to caregivers, follow-up appointments were difficult to obtain due to an inefficient referral system, and telemedicine was implemented only to a limited extent because of restricted internet access and limited technological literacy among patients and caregivers (63).

Policy and environmental level

There were eight studies that reported an analysis of nationwide estimated data regarding ischemic stroke frequency among general population, using secondary databases of healthcare systems registries. We observed a decrease in the age-standardized incidence rate of ischemic stroke from 60.28 in 1990 to 41.81 per 100,000 inhabitants in 2021 (75). Similar results obtained for disability-adjusted life years, YLD, and YLL related to ischemic stroke. However, during 2017, the age-standardized mortality rate was up to 853 per 100,000 inhabitants among individuals aged between 85 and 89 years old (76).

Preventive education is part of a key policy for stroke prevention. In this context, primary care interventions were developed using educational and digital tools which aimed to improve stroke knowledge and awareness in individuals with high cardiovascular risk. These interventions directly engaged individuals in the design and distribution of educational materials; however, it did not include clinical or impact evaluation of individuals after the intervention (77, 78).

There was only one study that reported environmental factors such as ambient particulate matter pollution, household air pollution from solid fuels, and lead exposure as the 3rd, 10th, and 13th, attributable causes to overall stroke, respectively (79). However, there were no studies that assessed impact of country policies' changes or implementations regarding ischemic stroke care.

Discussion

Summary of results

In our scoping review, we identified 119 primary studies of ischemic stroke in Peru, reflecting a gradual expansion of research activity over the past two decades. Most evidence came

from isolated observational studies. Aggregate epidemiological estimates including the proportion of undetermined etiology (30%) and the thrombolysis rate (2.9%) are broadly consistent with studies from the region (80, 81). The scientific value of this review lies in mapping the context-specific drivers behind those numbers and in identifying where the research outputs fail to cover the actual burden and geography of disease in Peru.

Many challenges were identified, such as an incomplete work-up to identify non-usual etiologies in individual ischemic stroke reports, the very low thrombolysis rate and the absence of reports on mechanical thrombectomy for acute care, the lack of clinical results for innovative computer-based interventions, the heterogeneous qualitative descriptions of patient and caregivers' experiences during post-stroke care and health system access limitations, and few reports about policy interventions and environmental risk factors. Overall, these findings reveal a research ecosystem that, despite meaningful growth, is partially aligned with Peru's growing ischemic stroke burden, focused on already known descriptive epidemiology.

Ischemic stroke epidemiology in Peru

Evidence regarding the epidemiology of ischemic stroke in Peru derives from local predominantly single-center studies with 79.1% originating from the capital city. Meanwhile, national-level estimates often depend on secondary data from healthcare systems, with reporting and methodological limitations (38, 82). The absence of population-based studies that consider Peru's cultural and geographical diversity likely resulted in an underestimation of the true burden of ischemic stroke and may bias policy decisions related to stroke prevention and care. Furthermore, analysis of ischemic stroke etiologies revealed that nearly one third of patients had an undetermined etiology, differing with a previous global meta-analysis (83). However, this result is within the range of other studies in Latin America (81), Brazil (84), Chile (85), Mexico (86). Thus, in high-income settings, cryptogenic stroke typically reflects residual uncertainty after exhaustive investigation, whereas in Peru and in Latin America, it may reflect a structural absence of the diagnostic tools required to complete that investigation, such as vascular imaging, prolonged cardiac monitoring, structural cardiac evaluations, or coagulopathy testing, contrary to recommendations from the 2021 AHA/ASA secondary prevention guidelines (87). These diagnostic gaps limit the identification of other specific etiologies with potential specific treatments to reduce the recurrence of ischemic stroke.

Regarding ischemic stroke risk factors, research findings in Peru align with well-established determinants such as high prevalences of hypertension, dyslipidemia, cardiovascular disease, hormonal contraceptive use, and aging (88). A few studies explored high blood viscosity in Andean populations living at high altitudes as a potential contributor for ischemic stroke (89, 90). Nonetheless, there was lack of information regarding other risk factors particularly relevant to the Peruvian context, including the high prevalence of intracranial arterial stenosis in native-Hispanic population, the effects of high-altitude residence, and the influence of socioeconomic conditions and cultural beliefs on chronic disease healthcare

access and treatment adherence (91, 92). On the other hand, some studies identified ischemic stroke as a risk factor for depression and cognitive impairment in Peruvian population (39, 40). However, research has not yet explored the long-term social implications of stroke, effects on sexual health, or economic burden for households and the health system. These research gaps show the need for integrative, context-sensitive research agendas that can inform equitable and culturally adapted stroke prevention strategies in low- and middle-income settings.

Evidence-based interventions for ischemic stroke in Peru

In Peru, interventional evidence remains scarce. While Peruvian patients participated in multicenter stroke prevention or therapeutic trials, the local and individualized data were unavailable. This is particularly relevant because the implementation and adherence of evidence-based recommendations for ischemic stroke in Peruvian centers were not achieved by a multifaceted quality improvement intervention during a randomized cluster trial (69). In parallel, local adherence to AHA/ASA measures has been suboptimal and thrombolysis rates historically low (9, 26). The thrombolysis rate ranged between 1.19% and 6.3% (9, 24, 31, 36, 93, 94), similar to rates reported in low- and middle-income countries around the world (95) and Latin America (81), like Brazil (96), while there are no reports of mechanical thrombectomy. This deficiency represents a system-level failure with directly actionable solution to multifactorial causes including delayed patient arrival outside the therapeutic window (9, 24), poor recognition of stroke symptoms (9, 36), lack of adequate pre-hospital notification and transport (9, 36), insufficient availability of CT and MRI scanners (24), and inadequate acute care flow (24). This is highlighted by a recently published study, not included in the present analysis, that reported an heterogeneous thrombolysis rate between 7.7% to 26.3% across five Peruvian facilities from 2023 to 2025, where the highest rate was in a private center from Lima (68). Furthermore, there is still an urgent need for contextual adaptation in service delivery of well-known evidence-based interventions for stroke treatment (97, 98).

To address this critical gap in acute care and adherence, we propose a contextually adapted framework for the Peruvian health system. This framework requires establishing a national program that standardizes Code Stroke pathways across all levels of care and assures complete access to evidence-based interventions. Structurally, this entails dividing the country into geographical zones (central, north, south, and east) where third-level hospitals equipped with specialized stroke units must be interconnected for rapid decision-making, based on the 2023 Telemedicine Technical Guideline to expand tele-stroke networks beyond Lima. Finally, this program must incorporate rehabilitation centers and secondary prevention at the primary and secondary levels, alongside quality monitoring and multicenter research networks, ensuring actions align with global recommendations and system guidance for middle-income nations (7, 99, 100).

Caregivers and rehabilitation in post-stroke patients in Peru

Findings from both quantitative and qualitative studies highlight the gaps in post-stroke care in Peru, particularly regarding assistance and training for caregivers, rehabilitation programs, and access to palliative care. The post-stroke care continuum is fragmented, marked by systemic barriers to healthcare access (11, 24, 101), including financial constraints, affecting low-income populations (102), the absence of structured palliative care training programs (60, 103), insufficient institutional support for caregivers (61, 63, 102). After hospital discharge, stroke survivors were frequently discharged to their home despite needing specialized nursing or palliative support, transferring the burden of care to family members who often lack both financial and technical resources (60, 63). This lack of continuity of care forces caregivers to assume complex roles without adequate technical or economic resources, severely interrupting the rehabilitation process.

This reality exposes profound disparities when compared with high-income countries, where the access to rehabilitation program is about 66% (104), higher compared to low- and middle-income countries settings, such as Peru where there is not a national palliative care hospital, program or policy (11, 63, 103). Strengthening post-stroke care in Peru and similar low- and middle-income countries requires the implementation of multicomponent interventions that integrate rehabilitation, palliative care, and psychological support. Such an approach should prioritize community-based delivery models and caregiver training to enhance independence in activities of daily living, reduce caregiver burden, and ultimately improve long-term outcomes after stroke.

Limitations and strengths

This scoping review is limited by substantial heterogeneity in study designs, populations, outcomes, and reporting, which precluded quantitative synthesis and limited direct comparability across studies. Also, the exclusion of cerebral venous thrombosis means that our estimates should be interpreted as representing arterial ischemic stroke rather than the broader burden of all ischemic cerebrovascular disease. This may be particularly relevant in younger women, given the association of cerebral venous thrombosis with hormonal exposure, pregnancy, and the puerperium (105). The classification following a socio-ecological approach involved some degree of subjective judgment, as determinants may overlap across socio-ecological levels despite the use of consensus-based classification. Also, there is missing information among studies for critical outcomes such as disability and mortality, and how these present in relation to other clinical and imaging characteristics. The literature was heavily concentrated in Lima and in facility-based secondary data, introducing geographic and selection biases and likely underestimating stroke burden in rural, Andean, and Amazonian populations. Publication bias (unpublished local

reports and grey literature) and the paucity of large and randomized trials further limit the generalizability of some findings. Finally, despite that most of the included reports on interventional rehabilitation technologies and community interventions came from conference abstracts lacking proper peer-review and validity for applicability, we included them because of the limited number of originally published interventional studies.

Scoping reviews addressing specific aspects of stroke care have been conducted in other Latin American countries^(106–108). However, we did not identify a directly comparable national evidence-mapping study integrating epidemiological, clinical, organizational, community, and policy evidence within a socio-ecological framework. This review provides the most comprehensive mapping to date of published empirical evidence on ischemic stroke in Peru. We used a sensitive search strategy to include and integrate different study designs including observational, interventional, qualitative, and case-report literature. Use of explicit scoping methodology and inclusion of implementation-relevant outcomes (timing, thrombolysis rates, caregiver experiences, and adherence to performance measures) enhances the review's utility for clinicians, health-system planners and researchers.

Conclusion

Evidence on ischemic stroke in Peru is concentrated in urban tertiary centers, heterogeneous, and limited by frequent incomplete etiologic workups. Aggregated epidemiological estimates are broadly consistent with regional reports, such as high proportions of undetermined etiology and very low reperfusion treatment access. However, the identified context-specific causes for these estimates were diagnostic infrastructure deficits, absence of acute care pathways, and fragmented post-stroke support. Major gaps persist in rural and high-altitude settings, rehabilitation, caregiver support, and palliative care. To improve outcomes, Peru will require a nationally coordinated, regionally distributed stroke program that standardizes acute pathways, strengthens diagnostic capacity, and scales community-based rehabilitation, caregiver training, mental-health, and palliative services.

Data availability statement

The original contributions presented in the study are included in the article/[Supplementary Material](#), further inquiries can be directed to the corresponding author.

Author contributions

CA-R: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Writing – original draft, Writing – review & editing. JG-A: Data curation, Investigation, Methodology, Writing – original draft, Writing –

review & editing. LC-Q: Data curation, Investigation, Methodology, Writing – original draft, Writing – review & editing. EB-V: Data curation, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing. JM-A: Data curation, Investigation, Methodology, Writing – original draft, Writing – review & editing. RO-S: Data curation, Investigation, Methodology, Writing – original draft, Writing – review & editing. RE: Data curation, Investigation, Methodology, Writing – original draft, Writing – review & editing. JZ: Investigation, Supervision, Writing – original draft, Writing – review & editing. JM: Investigation, Supervision, Writing – original draft, Writing – review & editing.

Funding

The author(s) declared that financial support was not received for this work and/or its publication.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

The handling editor APO declared a past co-authorship with the author(s) JJM.

Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

Any alternative text (alt text) provided alongside figures in this article has been generated by Frontiers with the support of artificial intelligence and reasonable efforts have been made to ensure accuracy, including review by the authors wherever possible. If you identify any issues, please contact us.

Publisher's note

All claims expressed in this article are solely those of the authors and do not necessarily represent those of their affiliated organizations, or those of the publisher, the editors and the reviewers. Any product that may be evaluated in this article, or claim that may be made by its manufacturer, is not guaranteed or endorsed by the publisher.

Supplementary material

The Supplementary Material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/fepid.2026.1926861/full#supplementary-material>

References

- Feigin VL, Brainin M, Norrving B, Martins SO, Pandian J, Lindsay P, et al. World stroke organization: global stroke fact sheet 2025. *Int J Stroke*. (2025) 20(2):132–44. doi: 10.1177/17474930241308142
- Feigin VL, Stark BA, Johnson CO, Roth GA, Bisignano C, Abady GG, et al. Global, regional, and national burden of stroke and its risk factors, 1990–2019: a systematic analysis for the global burden of disease study 2019. *Lancet Neurol*. (2021) 20(10):795–820. doi: 10.1016/S1474-4422(21)00252-0
- Rodríguez DMG, Grandez JDP, Dianderas FV, Sara JDC. Incidence of cerebrovascular disease in Peru from 2015 to 2023. *PLOS Glob Public Health*. (2025) 5(5):e0004559. doi: 10.1371/journal.pgph.0004559
- Majumder D. Ischemic stroke: pathophysiology and evolving treatment approaches. *Neurosci Insights*. (2024) 19:26331055241292600. doi: 10.1177/26331055241292600
- Li X-, Kong X-, Yang C-, Cheng Z-, Lv J-, Guo H, et al. Global, regional, and national burden of ischemic stroke, 1990–2021: an analysis of data from the global burden of disease study 2021. *eClinicalMed*. (2024) 75:102758. doi: 10.1016/j.eclinm.2024.102758
- Cagna-Castillo D, Salcedo-Carrillo AL, Carrillo-Larco RM, Bernabé-Ortiz A. Prevalence and incidence of stroke in Latin America and the Caribbean: a systematic review and meta-analysis. *Sci Rep*. (2023) 13(1):6809. doi: 10.1038/s41598-023-33182-3
- Feigin VL, Owolabi MO, Feigin VL, Abd-Allah F, Akinyemi RO, Bhattacherjee NV, et al. Pragmatic solutions to reduce the global burden of stroke: a world stroke organization–lancet neurology commission. *Lancet Neurol*. (2023) 22(12):1160–206. doi: 10.1016/S1474-4422(23)00277-6
- Martins SCO, Lavados P, Secchi TL, Brainin M, Ameriso S, Gongora-Rivera F, et al. Fighting against stroke in Latin America: a joint effort of medical professional societies and governments. *Front Neurol*. (2021) 12:743732. doi: 10.3389/fneur.2021.743732
- Abanto C, Valencia A, Calle P, Barrientos D, Flores N, Novoa M, et al. Challenges of thrombolysis in a developing country: characteristics and outcomes in Peru. *J Stroke Cerebrovasc Dis*. (2020) 29(7):104819. doi: 10.1016/j.jstrokecerebrovasdis.2020.104819
- Ramírez Preciado MC, Apolinar Gonzáles N, Zavaleta Llanos EW, Palomino Márquez MG, Correa Huanan D. Access to health in the periphery of Peru: a systematic review. *Rev InveCom*. (2026) 6(3). doi: 10.5281/zenodo.17372482
- Miranda JJ, Moscoso MG, Yan LL, Diez-Canseco F, Málaga G, García HH, et al. Addressing post-stroke care in rural areas with Peru as a case study. Placing emphasis on evidence-based pragmatism. *J Neurol Sci*. (2017) 375:309–15. doi: 10.1016/j.jns.2017.02.027
- Almubaslat F, Sanchez-Boluarte SS, Diaz MM. A review of neurological health disparities in Peru. *Front Public Health*. (2023) 11:1210238. doi: 10.3389/fpubh.2023.1210238
- Whitehorn A, Fu L, Porritt K, Lizarrondo L, Stephenson M, Marin T, et al. Mapping clinical barriers and evidence-based implementation strategies in low-to-middle income countries (LMICs). *Worldviews Evid Based Nurs*. (2021) 18(3):190–200. doi: 10.1111/wvn.12503
- Cuevas C, Kikushima-Alcantara P, Miranda JJ. The state of cardiovascular care in Peru: needs for system strengthening. *Circulation*. (2025) 152(1):7–10. doi: 10.1161/CIRCULATIONAHA.124.071776
- Cooke A, Smith D, Booth A. Beyond PICO: the SPIDER tool for qualitative evidence synthesis. *Qual Health Res*. (2012) 22(10):1435–43. doi: 10.1177/1049732312452938
- Baatiema L, Sanuade O, Kuumuori Ganle J, Sumah A, Baatiema L, Sumankuuro J. An ecological approach to understanding stroke experience and access to rehabilitation services in Ghana: a cross-sectional study. *Health Soc Care Community*. (2021) 29(5):e67–78. doi: 10.1111/hsc.13243
- Yang M, Hu Z, Qu X, Li M, Du W, Fan L. Utilization and effectiveness of the PRECEDE-PROCEED model as a tool in public health interventions: a systematic review. *Iran J Public Health*. (2025) 54(10):2073–88. doi: 10.18502/ijph.v54i10.20121
- Richards CT. Strengthening the stroke chain of survival in community emergency departments. *J Am Coll Emerg Physicians Open*. (2022) 3(4):e12763. doi: 10.1002/emp2.12763
- Adams HP, Bendixen BH, Kappelle LJ, Biller J, Love BB, Gordon DL, et al. Classification of subtype of acute ischemic stroke. Definitions for use in a multicenter clinical trial. TOAST. Trial of org 10172 in acute stroke treatment. *Stroke*. (1993) 24(1):35–41. doi: 10.1161/01.STR.24.1.35
- Moyano LM, Montano SM, Vilchez Barreto P, Reto N, Larrauri L, Mori N, et al. Prevalence of stroke survival in rural communities living in northern Peru. *PLoS One*. (2021) 16(7):e0254440. doi: 10.1371/journal.pone.0254440
- Llibre Rodríguez JJ, Prina AM, Acosta D, Guerra M, Huang Y, Jacob KS, et al. The prevalence and correlates of frailty in urban and rural populations in Latin America, China, and India: a 10/66 population-based survey. *J Am Med Dir Assoc*. (2018) 19(4):287–295.e4. doi: 10.1016/j.jamda.2017.09.026
- Huamani C, Oré-Montalvo V, Bayona-Pancorbo W, Pérez-Alviz C, Acuña-Mamani JC, Córdova-Heredia G, et al. Mortalidad y pronóstico funcional en pacientes con infarto cerebral en una ciudad a gran altitud (Cusco, 3300 m) durante la pandemia por COVID-19. *Rev Ecuat Neurol*. (2024) 33(2):21–7. doi: 10.46997/reveuatneuro133200021
- Enciso Matos I, Pinto-Casaverde L, Calle La Rosa P, Torres-Ramírez L. Etiología de infarto cerebral en adultos jóvenes de una serie de pacientes del instituto nacional de ciencias neurológicas en el periodo 2010 al 2013. *Rev Neuro-Psiquiatr*. (2016) 79(1):16. doi: 10.20453/rnp.v79i1.2764
- Rodríguez-Kadota L, Vences MA, Failoc-Rojas VE. Eventos cerebro vasculares en pacientes COVID-19: experiencia en un hospital de referencia nacional del Perú. *Rev Cuerpo Méd Hosp Nac Almanzor Aguinaga Asenjo*. (2021) 14(3):357–61. doi: 10.35434/rcmhnaaa.2021.143.1257
- Alvarado-Dulanto CMA, Lazo M, De Los Á, Loza-Herrera JD, Málaga G. Pronóstico al año tras sufrir el debut de enfermedad cerebrovascular en pacientes de un hospital nacional de Lima, Perú. *Rev Peru Med Exp Salud Pública*. (2015) 32(1):98–103. doi: 10.17843/rpmesp.2015.32.1581
- Abanto C, Ulrich AK, Valencia A, Dueñas V, Montano S, Tirschwell D, et al. Adherence to American Heart Association/American stroke association clinical performance measures in a Peruvian neurological reference institute. *J Stroke Cerebrovasc Dis*. (2020) 29(11):105285. doi: 10.1016/j.jstrokecerebrovasdis.2020.105285
- Dittrich LB, Beck da Silva Etges AP, Siqueira de Souza J, Zago Marcolino MA, Rocha E, Amaya P, et al. Cost evaluation of acute ischemic stroke in Latin America: a multicentric study. *Lancet Reg Health—Am*. (2025) 41:100959. doi: 10.1016/j.lana.2024.100959
- Abanto C, Ton TGN, Tirschwell DL, Montano S, Quispe Y, Gonzales I, et al. Predictors of functional outcome among stroke patients in Lima, Peru. *J Stroke Cerebrovasc Dis*. (2013) 22(7):1156–62. doi: 10.1016/j.jstrokecerebrovasdis.2012.11.021
- Coronado Munoz A, Tasayco J, Morales W, Moreno L, Zorrilla D, Stapleton A, et al. High incidence of stroke and mortality in pediatric critical care patients with COVID-19 in Peru. *Pediatr Res*. (2022) 91(7):1730–4. doi: 10.1038/s41390-021-01547-x
- Salazar-Varela YK, Segura-Plasencia NM, Montalvan-Yparraguirre VM. Relación entre uricemia Y el estado funcional neurológico en el Ictus isquémico agudo— estudio multicéntrico. *Rev Ecuat Neurol*. (2020) 29(3):49–54. doi: 10.46997/reveuatneuro129300049
- Vences MA, Failoc-Rojas VE, Urrunaga-Pastor D, Hurtado-Roca Y. Risk factors for in-hospital complications in patients with acute ischemic stroke: retrospective cohort in a national reference hospital in Peru. *Heliyon*. (2023) 9(5):e15810. doi: 10.1016/j.heliyon.2023.e15810
- Tirado AG. Determinación del riesgo de muerte por afecciones vasculares en hipertensos de la población peruana 2021–2022. *Rev Virtual Soc Paraguaya Med Interna*. (2023) 10:12–20. doi: 10.18004/rvpsmi/2312-3893/2023.10.02.12
- Tarazona B, Ramos W, Arce J, Yarinsueca J, Morales S, Ronceros G, et al. Etiology and risk factors for a first episode of cerebral ischemia in young adults. *Neurol Engl Ed*. (2010) 25(8):470–7. doi: 10.1016/S2173-5808(10)70088-2
- Asmat-Vásquez JCM, Díaz-Paz K. Índice leucoglucémico alto como predictor de mortalidad intrahospitalaria en pacientes con ictus isquémico agudo. *Rev Ecuat Neurol*. (2021) 30(2):18–25. doi: 10.46997/reveuatneuro130200018
- Sanchez-Boluarte SS, Bejarano-Ferreira J, Lescano W, Valdez-Taboada ME, Barrientos-Iman DM, García HH. Resultado funcional en pacientes con infarto cerebral y COVID-19 en Lima, Perú. *Neurol Argent*. (2023) 15(2):87–92. doi: 10.1016/j.neuarg.2023.03.003
- Gallo-Guerrero M, Zevallos C, Quiñónez M. Factores asociados a resultados funcionales en pacientes con ictus isquémico tratados con trombolisis endovenosa en un hospital del Perú. *Rev Neuro-Psiquiatr RNP*. (2020) 83(2):79–86. doi: 10.20453/rnp.v83i2.3750
- Castilla-Torres NV, Díaz-Tavera Z, Condorhuanan-Figueroa Y, Ango-Bedriñana J, Ramirez JK. Clinical and pharmacotherapeutic factors as survival and death predictors in hospitalized post-stroke patients. *Neurosci Res Notes*. (2023) 6(2):180.1–9. doi: 10.31117/neuroscirn.v6i2.180
- Labán-Seminario LM, Carrillo-Larco RM, Bernabé-Ortiz A. Stroke-related length of hospitalization trends and in-hospital mortality in Peru. *PeerJ*. (2022) 10:e14467. doi: 10.7717/peerj.14467
- Herrera M, Lama J, De-la-Cruz J. Factores de riesgo de depresión post ictus isquémico. *Rev Médica Hered*. (2020) 31(3):181–9. doi: 10.20453/rmh.v31i3.3808
- Custodio N, Montesinos R, Alva-Díaz C, Pacheco-Barrios K, Rodríguez-Calienes A, Herrera-Pérez E, et al. Diagnostic accuracy of brief cognitive screening tools to diagnose vascular cognitive impairment in Peru. *Int J Geriatr Psychiatry*. (2022) 37(1):1–10. doi: 10.1002/gps.5531
- Yrribery Ureña S, Monge Zapata V, Salazar Cabrera F, Barriga Calle E, Vesco Montegudo E. Gastrotomía endoscópica percutánea: experiencia prospectiva de

- un centro privado nacional. *Rev Gastroenterol Perú*. (2004) 24(4):314–22. Available online at: <https://revistagastroperu.com/index.php/rgp/article/view/700> (Accessed February 15, 2026).
42. Achanccaray D, Flores C, Fonseca C, Andreu-Perez J. A P300-based brain computer interface for smart home interaction through an ANFIS ensemble. In: *2017 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*, 2017, 1–5. doi: 10.1109/FUZZ-IEEE.2017.8015770
 43. Achanccaray D, Flores C, Fonseca C, Andreu-Perez J. A fuzzy genetic algorithm for optimal spatial filter selection for P300-based brain computer interfaces. In: *2018 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)* (2018), 1–6. doi: 10.1109/FUZZ-IEEE.2018.8491676
 44. Cortez SA, Flores C, Andreu-Perez J. Single-trial P300 classification using deep belief networks for a BCI system. In: *2020 IEEE XXVII International Conference on Electronics, Electrical Engineering and Computing (INTERCON)* (2020), 1–4. doi: 10.1109/INTERCON50315.2020.9220255
 45. Cortez SA, Flores C, Andreu-Perez J. Improving speller BCI performance using a cluster-based under-sampling method. In: *2020 IEEE Symposium Series on Computational Intelligence (SSCI)* (2020), 576–81. doi: 10.1109/SSCI47803.2020.9308174
 46. Cortez SA, Flores C, Andreu-Perez J. Under-sampling and classification of P300 single-trials using self-organized maps and deep neural networks for a speller BCI. In: *2020 IEEE International Conference on Systems, Man, and Cybernetics (SMC)* (2020), 2972–8. doi: 10.1109/SMC42975.2020.9283178
 47. Cortez SA, Flores C, Andreu-Perez J. A smart home control prototype using a P300-based brain-computer interface for post-stroke patients. In: Iano Y, Arthur R, Saotome O, Kemper G, Borges Monteiro AC, editors. *Proceedings of the 5th Brazilian Technology Symposium*. Springer International Publishing (2021) p. 131–9. doi: 10.1007/978-3-030-57566-3_13
 48. Flores C, Fonseca C, Achanccaray D, Andreu-Perez J. Performance evaluation of a P300 brain-computer interface using a kernel extreme learning machine classifier. In: *2018 IEEE International Conference on Systems, Man, and Cybernetics (SMC)* (2018), 3715–9. doi: 10.1109/SMC.2018.00629
 49. Vega CF, Quevedo J, Escandón E, Kiani M, Ding W, Andreu-Perez J. Fuzzy temporal convolutional neural networks in P300-based brain-computer interface for smart home interaction. *Appl Soft Comput*. (2022) 117:108359. doi: 10.1016/j.asoc.2021.108359
 50. Chirinos SC, Vela EA. A modular soft robotic exoskeleton for active hand rehabilitation after stroke. In: *2021 IEEE Engineering International Research Conference (EIRCON)* (2021), 1–4. doi: 10.1109/EIRCON52903.2021.9613403
 51. Huamanchahua D, Sierra-Huertás J, Terrazas-Rodas D, Janampa-Espinoza A, Gonzáles J, Huamán-Vizconde S. Kinematic analysis of an 4 DOF upper-limb exoskeleton. In: *2021 IEEE 12th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON)* (2021), 0914–23. doi: 10.1109/UEMCON53757.2021.9666604
 52. Nuñez-Quispe J, Figueroa A, Campusano D, Huamanchumo J, Soto A, Chate E, et al. Structural design and sliding mode control approach of a 4-DOF upper-limb exoskeleton for post-stroke rehabilitation. In: Pucheta M, Cardona A, Preidikman S, Hecker R, editors. *Multibody Mechatronic Systems*. Springer International Publishing (2022) p. 213–23. doi: 10.1007/978-3-030-88751-3_22
 53. Vargas M, Mayorga J, Osco B, Cuyotupac V, Nacarino A, Allcca D, et al. Bio-Mechatronics development of robotic exoskeleton system with Mobile-prismatic joint mechanism for passive hand wearable-rehabilitation. *Emerg Sci J*. (2024) 8(6):2144–72. doi: 10.28991/ESJ-2024-08-06-02
 54. Cervantes CM, Sandoval C, Palomares R, Arroyo JB, Manrique MM, Cornejo J. Evaluación antropométrica biomédica y diseño mecánico conceptual de un sistema robótico para la rehabilitación pasiva de miembros inferiores en pacientes post-accidente cerebrovascular. *Rev Fac Med Humana*. (2024) 24(2):72–81. doi: 10.25176/rfmh.v24i2.6550
 55. Achanccaray D, Acuña K, Carranza E, Andreu-Perez J. A virtual reality and brain computer interface system for upper limb rehabilitation of post stroke patients. In: *2017 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)* (2017), 1–5. doi: 10.1109/FUZZ-IEEE.2017.8015726
 56. Achanccaray D, Pacheco K, Carranza E, Hayashibe M. Immersive virtual reality feedback in a brain computer interface for upper limb rehabilitation. In: *2018 IEEE International Conference on Systems, Man, and Cybernetics (SMC)* (2018), 1006–10. doi: 10.1109/SMC.2018.00179
 57. Oroya J, Oroya D, Huamanchahua D. Design a hand orthosis to aid post-stroke patients with hemiplegia. In: *2021 IEEE 12th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON)* (2021), 0752–7. doi: 10.1109/UEMCON53757.2021.9666638
 58. Casas EL, Callupe RL, Elías D. A telerehabilitation web-based system for remote evaluation of patients with gait pathology. In: *2010 Pan American Health Care Exchanges* (2010), 112. doi: 10.1109/PAHCE.2010.5474587
 59. Poole JE, Casas JPC, Bolli RA, Krebs HI. A real-time algorithm for identifying gait in a sensory substitution system. In: *2019 IEEE MIT Undergraduate Research Technology Conference (URTC)* (2019), 1–4. doi: 10.1109/URTC49097.2019.9660511
 60. Abedini NC, Shulein OM, Berrú-Villalobos S, Ramírez-Quinones J, Creutzfeldt CJ, Portocarrero J, et al. Outcomes and experiences of patients and their caregivers after severe stroke requiring tube feeding in Peru. *J Pain Symptom Manage*. (2024) 67(4):296–305. doi: 10.1016/j.jpainsymman.2024.01.001
 61. Pesantes MA, Brandt LR, Ipince A, Miranda JJ, Diez-Canseco F. An exploration into caring for a stroke-survivor in Lima, Peru: emotional impact, stress factors, coping mechanisms and unmet needs of informal caregivers. *eNeurologicalSci*. (2017) 6:33–50. doi: 10.1016/j.ensci.2016.11.004
 62. Asencios Trujillo L, Asencios Trujillo LV, La Rosa Longobardi CJ, Gallegos Espinoza D, Piñas Rivera LC. Level of caregiver overload in patients diagnosed with stroke in a specialized hospital institution in metropolitan Lima. *Health Leadersh Qual Life*. (2023) 2:25. doi: 10.56294/hl202325
 63. Zafra-Tanaka JH, Portocarrero J, Abanto C, Zunt JR, Miranda JJ. Managing post-stroke care during the COVID-19 pandemic at a tertiary care level hospital in Peru. *J Stroke Cerebrovasc Dis*. (2022) 31(4):106275. doi: 10.1016/j.jstrokecerebrovasdis.2021.106275
 64. McDonald CL, Fuhs AK, Kartin D. An examination of current stroke rehabilitation practice in Peru: implications for interprofessional education. *J Interprof Care*. (2018) 32(3):329–38. doi: 10.1080/13561820.2018.1425290
 65. Lira-Mamani D, Concha-Flores G. Factors affecting the arrival time of cerebrovascular disease patients at the emergency department of a Peruvian hospital. *Rev Neurol*. (2004) 39(6):508–12. Available online at: <https://pubmed.ncbi.nlm.nih.gov/15467985/> (Accessed February 15, 2026).
 66. Valenzuela-Rodríguez G, Lescano-Alva M, Bryce-Alberti M, Portmann-Baracco A, Prudencio-León W. Complicaciones cardiovasculares agudas en una población peruana de pacientes oncológicos. *Arch Peru Cardiol Cir Cardiovasc*. (2022) 3(1):1–7. doi: 10.47487/apcyccv.v3i1.192
 67. Saal-Zapata G, Rodríguez R. Trombectomía mecánica, aspirativa e intra-arterial. Primer caso en un paciente con stroke isquémico agudo de un hospital nacional. *Rev Médica Hered*. (2020) 31(1):37–41. doi: 10.20453/rmh.v3i1i.3726
 68. Gallo-Guerrero M, Aguirre-Quispe W, Gallo-Lazarte CN, Sanchez-Boluarte SS, Peralta-Mestas A, Kadota LER, et al. Real-world characteristics and outcomes of ischemic stroke: a Peruvian multicenter analysis from the RES-Q registry. *Front Stroke*. (2026) 5:1783830. doi: 10.3389/fstro.2026.1783830
 69. Machline-Carrion MJ, Santucci EV, Damiani LP, Bahit MC, Málaga G, Pontes-Neto OM, et al. Effect of a quality improvement intervention on adherence to therapies for patients with acute ischemic stroke and transient ischemic attack: a cluster randomized clinical trial. *JAMA Neurol*. (2019) 76(8):932–41. doi: 10.1001/jamaneurol.2019.1012
 70. Collantes Cubas JA, Pérez Ventura SA, Vigil-De Gracia P, Castañeda Bazán KE, Tapia Saldaña JM, Leyva FJ. Maternal mortality in pregnant women with positive SARS-CoV-2 antibodies and severe preeclampsia. Report of 3 cases. *Rev Peru Ginecol Obstet*. (2020) 66(3):00013. doi: 10.31403/rpgo.v66i2279
 71. Mariños E, Barreto-Acevedo E, Espino P. Accidente cerebrovascular isquémico asociado a COVID-19: primer reporte de casos en Perú. *Rev Neuro-Psiquiatr*. (2020) 83(2):127–33. doi: 10.20453/rnp.v83i2.3756
 72. Caballero-Alvarado J, Millones KR, Gonzales MR, Alvarado ABR, Corvera CZ, Barboza JJ. Ischemic stroke associated with multisystemic inflammatory syndrome in children after SARS-CoV-2 infection: a case report. *Pediatr Med*. (2023) 6:40. doi: 10.21037/pm-22-43
 73. Nogueira RG, Qureshi MM, Abdalkader M, Martins SO, Yamagami H, Qiu Z, et al. Global impact of COVID-19 on stroke care and IV thrombolysis. *Neurology*. (2021) 96(23):e2824–38. doi: 10.1212/WNL.00000000000011885
 74. Pujol-Lereis VA, Flores A, Barboza MA, Abanto-Argomedo C, Amaya P, Bayona H, et al. COVID-19 Lockdown effects on acute stroke care in Latin America. *J Stroke Cerebrovasc Dis*. (2021) 30(9):105985. doi: 10.1016/j.jstrokecerebrovasdis.2021.105985
 75. Feigin VL, Abate MD, Abate YH, Abd ElHafeez S, Abd-Allah F, Abdelalim A, et al. Global, regional, and national burden of stroke and its risk factors, 1990–2021: a systematic analysis for the global burden of disease study 2021. *Lancet Neurol*. (2024) 23(10):973–1003. doi: 10.1016/S1474-4422(24)00369-7
 76. Roth GA, Abate D, Abate KH, Abay SM, Abbafati C, Abbasi N, et al. Global, regional, and national age-sex-specific mortality for 282 causes of death in 195 countries and territories, 1980–2017: a systematic analysis for the global burden of disease study 2017. *Lancet*. (2018) 392(10159):1736–88. doi: 10.1016/S0140-6736(18)32203-7
 77. Turk K, Zunt J, Abanto C, Sanchez A. *Development of a WhatsApp-Based Pilot Program to Improve Stroke Education and Risk Factor Reduction in Patients with Arterial Hypertension in Huaral, Peru*. CUGH (2022). Available online at: <https://cugh.confex.com/cugh/2022/meetingapp.cgi/Paper/1956> (Accessed November 3, 2025).
 78. Tulloch-Palomino MV. Improving stroke awareness in urban and rural Peru through a hospital-based stroke prevention campaign. *J Invest Med*. (2016) 64:207–8. doi: 10.1136/jim-d-15-00013.167
 79. Pacheco-Barrios K, Giannoni-Luza S, Navarro-Flores A, Rebello-Sanchez I, Parente J, Balbuena A, et al. Burden of stroke and population-attributable fractions of risk factors in Latin America and the Caribbean. *J Am Heart Assoc*. (2022) 11(21):e027044. doi: 10.1161/JAHA.122.027044
 80. Alonso de Leciana M, Mazya MV, Kostulas N, Del Brutto OH, Abanto C, Massaro AR, et al. Stroke care and application of thrombolysis in ibero-America:

- report from the SITS-SIECV ibero-American stroke register. *Stroke*. (2019) 50(9):2507–12. doi: 10.1161/STROKEAHA.119.025668
81. Cantú-Brito C, Sampaio Silva G, Ameriso SF. Embolic stroke of undetermined source in Latin America: a review. *Neurologist*. (2017) 22(5):171–81. doi: 10.1097/NRL.0000000000000128
82. Bernabé-Ortiz A, Carrillo-Larco RM. Tasa de incidencia del accidente cerebrovascular en el Perú. *Rev Peru Med Exp Salud Pública*. (2021) 38:399–405. doi: 10.17843/rpmesp.2021.383.7804
83. Ornello R, Degan D, Tiseo C, Di Carmine C, Perciballi L, Pistoia F, et al. Distribution and temporal trends from 1993 to 2015 of ischemic stroke subtypes: a systematic review and meta-analysis. *Stroke*. (2018) 49(4):814–9. doi: 10.1161/STROKEAHA.117.020031
84. Rocha LJA, Silva KA, Chagas AL, Veras AO, Souto VGL, Valente MCMB, et al. Stroke in the state of Alagoas, Brazil: a descriptive analysis of a northeastern scenario. *Arq Neuropsiquiatr*. (2022) 80(6):550–6. doi: 10.1590/0004-282X-ANP-2021-0194
85. Vallejos J, Jaramillo A, Reyes A, Illanes S, Orellana P, Manterola J, et al. Prognosis of cryptogenic ischemic stroke: a prospective single-center study in Chile. *J Stroke Cerebrovasc Dis*. (2012) 21(8):621–8. doi: 10.1016/j.jstrokecerebrovasdis.2011.02.005
86. Cantú-Brito C, Ruiz-Sandoval JL, Chiquete E, Arauz A, León-Jiménez C, Murillo-Bonilla LM. Risk factors, causes and prognosis of cerebrovascular disease types in Mexico: RENAMEVASC study. *Rev Mex Neuroci*. (2011) 12(5):224–34. Available online at: <https://previous.revmexneurociencia.com/wp-content/uploads/2013/10/Nm115-02.pdf> (Accessed February 15, 2026).
87. Kleindorfer DO, Towfighi A, Chaturvedi S, Cockcroft KM, Gutierrez J, Lombardi-Hill D, et al. 2021 Guideline for the prevention of stroke in patients with stroke and transient ischemic attack: a guideline from the American Heart Association/American stroke association. *Stroke*. (2021) 52(7):e364–467. doi: 10.1161/STR.0000000000000375
88. Boehme AK, Esenwa C, Elkind MSV. Stroke risk factors, genetics, and prevention. *Circ Res*. (2017) 120(3):472–95. doi: 10.1161/CIRCRESAHA.116.308398
89. Huamani C, Oré-Montalvo V, Bayona-Pancorbo W, Pérez-Alviz C, Acuña-Mamani JC, Córdova-Heredia G, et al. High blood viscosity in patients with ischemic stroke residing at high altitude. *Rev Ecuat Neurol*. (2024) 33(1):41–6. doi: 10.46997/revneurologia33100041
90. Huamani C, Oré-Montalvo V, Bayona-Pancorbo W, Pérez-Alviz C, Acuña-Mamani J, Córdova-Heredia G, et al. Radiodensity of the middle cerebral artery and correlation with the hemoglobin level in patients with and without stroke living at high altitudes. *An Fac Med*. (2023) 84(3):267–71. doi: 10.15381/anales.v84i3.25464
91. Lazo-Porras M, Bernabe-Ortiz A, Gilman RH, Checkley W, Smeeth L, Miranda JJ. Population-based stroke incidence estimates in Peru: exploratory results from the CRONICAS cohort study. *Lancet Reg Health Am*. (2022) 5:100083. doi: 10.1016/j.lana.2021.100083
92. Del Brutto OH, Mera RM, Lama J, Zambrano M, Del Brutto VJ. Intracranial arterial stenosis in Ecuadorian natives/mestizos. A population-based study in older adults (the atahualpa project). *Arch Gerontol Geriatr*. (2015) 61(3):480–3. doi: 10.1016/j.archger.2015.08.011
93. Kawano-Castillo J, Chuquilín-Arista M, Tipismana-Barbarán M, Vizcarra-Escobar D. Factores asociados a la demora del tratamiento hospitalario de los pacientes con enfermedad cerebrovascular aguda. *Rev Neurol*. (2007) 44(5):264–8. doi: 10.33588/rn.4405.2006362
94. Gallo-Guerrero M, Vences MA, Zafra-Tanaka JH, Galindo D, Saavedra M, Zevallos C. Cerebrovascular disease and mortality in hospitalized patients with COVID-19 in a Latin American country, Peru. *Rev Neuro-Psiquiatr*. (2022) 85(3):194–205. doi: 10.20453/rnp.v85i3.4329
95. Saini V, Guada L, Yavagal DR. Global epidemiology of stroke and access to acute ischemic stroke interventions. *Neurology*. (2021) 97(20_Supplement_2):S6–S16. doi: 10.1212/WNL.0000000000012781
96. Goulart TO, Marazzi TBM, Monteiro RA, Camilo MR, Pontes-Neto OM. Ischemic stroke in Brazil (2017–2025): trends in admissions, thrombolysis, craniectomy, costs, and COVID-19 disruption. *J Stroke Cerebrovasc Dis*. (2026) 35(6):108631. doi: 10.1016/j.jstrokecerebrovasdis.2026.108631
97. Nguyen TH, Truong ALT, Ngo MB, Bui CTQ, Dinh QV, Doan TC, et al. Patients with thrombolysed stroke in Vietnam have an excellent outcome: results from the Vietnam thrombolysis registry. *Eur J Neurol*. (2010) 17(9):1188–92. doi: 10.1111/j.1468-1331.2010.02995.x
98. Nikkha K, Avan A, Shoeibi A, Azarpazhooh A, Ghandehari K, Foerch C, et al. Gaps and hurdles deter against following stroke guidelines for thrombolytic therapy in Iran: exploring the problem. *J Stroke Cerebrovasc Dis*. (2015) 24(2):408–15. doi: 10.1016/j.jstrokecerebrovasdis.2014.09.012
99. Ministerio de Salud. Guía técnica para el diagnóstico e inicio temprano del tratamiento de las personas con enfermedad cerebrovascular en la fase aguda a través de la telemedicina (2023). Available online at: <https://bvs.minsa.gob.pe/local/MINSA/6057.pdf> (Accessed November 3, 2025).
100. Pandian JD, Kalkonde Y, Sebastian IA, Felix C, Urimubenshi G, Bosch J. Stroke systems of care in low-income and middle-income countries: challenges and opportunities. *Lancet*. (2020) 396(10260):1443–51. doi: 10.1016/S0140-6736(20)31374-X
101. O'Donnell MJ, Chin SL, Rangarajan S, Xavier D, Liu L, Zhang H, et al. Global and regional effects of potentially modifiable risk factors associated with acute stroke in 32 countries (INTERSTROKE): a case-control study. *Lancet*. (2016) 388(10046):761–75. doi: 10.1016/S0140-6736(16)30506-2
102. Bernabe-Ortiz A, Diez-Canseco F, Vásquez A, Miranda JJ. Disability, caregiver's dependency and patterns of access to rehabilitation care: results from a national representative study in Peru. *Disabil Rehabil*. (2016) 38(6):582–8. doi: 10.3109/09638288.2015.1051246
103. Ecos-Quispe RL. *Feasibility and acceptability of an innovative hospital-based educational training for caregivers of patients with severe stroke at the instituto nacional de ciencias neurológicas in Perú in 2023* (Phd thesis). University of Washington Libraries (2024). Available online at: <https://digital.lib.washington.edu/researchworks/bitstreams/490710e9-db7a-4222-9b71-4fda2a12949b/download> (Accessed November 3, 2025).
104. Holloway RG, Arnold RM, Creutzfeldt CJ, Lewis EF, Lutz BJ, McCann RM, et al. Palliative and End-of-life care in stroke. *Stroke*. (2014) 45(6):1887–916. doi: 10.1161/STR.0000000000000015
105. Saposnik G, Bushnell C, Coutinho JM, Field TS, Furie KL, Galadanci N, et al. Diagnosis and management of cerebral venous thrombosis: a scientific statement from the American Heart Association. *Stroke*. (2024) 55(3):e77–90. doi: 10.1161/STR.0000000000000456
106. Vargas-Murcia JD, Isaza-Jaramillo SP, Uribe-Uribe CS. Risk factors and causes of ischemic stroke in young patients (18–49 years) in Colombia. A systematic review. *Rev Chil Neuro-Psiquiatr*. (2021) 59(2):113–24. doi: 10.4067/s0717-92272021000200113
107. Lima E, Teixeira-Salmela LF, Simões L, Guerra AC, Lemos A. Assessment of the measurement properties of the post stroke motor function instruments available in Brazil: a systematic review. *Braz J Phys Ther*. (2016) 20(2):114–25. doi: 10.1590/bjpt-rbf.2014.0144
108. Alcocer L. Challenges and treatment for stroke prophylaxis in patients with atrial fibrillation in Mexico: a review. *Am J Cardiovasc Drugs*. (2016) 16(3):171–82. doi: 10.1007/s40256-016-0163-6